Lecture 5.2: Data Collection via APIs

AREC 570 / ECON 530 / Unit 5

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Reading and sequence

Dominitz and Manski, "More Data or Better Data? A Statistical Decision Problem" (Dominitz and Manski 2017)

1. Survey the APIs available to economists
2. Retrieve four types of data in R
3. Check what each downloaded observation represents
4. Document a reproducible retrieval
5. Share the current data plan for each research project

Readings and course handouts are available on Canvas

APIs deliver secondary data

The organization responsible for the source determines

- What is measured
- Which people, places, or activities are covered
- When observations are collected and released
- How uncertainty, missing values, and revisions are recorded

An **application programming interface** provides a documented way for R to request selected records from that source

What happens when R calls an API

$$\text{R function} \rightarrow \text{request} \rightarrow \text{data service} \rightarrow \text{response} \rightarrow \text{R object}$$

- **Endpoint** is the web address for a data service
- **Parameters** specify the place, time, variables, and other selections
- **Response** is the information returned by the service
- **R package** translates the response into a familiar object

Many services transmit a structured text format called **JSON**. The packages in today's examples read that format and return a data frame, map, or raster

Getting access

Services use one of three arrangements

- **Open:** no registration, as with USGS Water Data, NASA POWER, and the World Bank
- **Agency key:** a free key from one agency, as with Census, BLS, BEA, FRED, and NASS
- **Shared key:** one api.data.gov key works across many federal services, including ERS and FoodData Central

A key is a personal credential. Store it in your user environment file and keep it out of shared code and repositories

The API landscape

Provider	Typical research uses
Census Bureau	Population, households, businesses, labor markets, trade
Other federal statistical agencies	Prices, income, employment, energy, finance
USDA	Production, prices, farm finances, food, soils, forests
Environment and natural resources	Water, weather, climate, land cover, pollution, hazards
International organizations	Cross-country development, agriculture, trade

As we scan the next slides, write down the two sources closest to your project

Census Bureau: people and places

Source	Economists use it for	R	Key
American Community Survey	Income, poverty, housing, and commuting by tract or county	`tidycensus`	Yes
ACS Public Use Microdata	Person and household records	`tidycensus`	Yes
Decennial Census	Population counts and demographics	`tidycensus`	Yes
Population Estimates	Annual population, births, deaths, and migration	`tidycensus`	Yes
SAIPE and SAHIE	County poverty, median income, and uninsured rates	`censusapi`	Yes

Census API dataset list | Census developer guidance

Census Bureau: businesses and labor markets

Source	Economists use it for	R	Key
County and ZIP Business Patterns	Establishments, employment, and payroll by industry	`censusapi`	Yes
Nonemployer Statistics	Self-employment and small businesses by industry	`censusapi`	Yes
Economic Census and Annual Business Survey	Sales, ownership, and firm characteristics	`censusapi`	Yes
Quarterly Workforce Indicators	Hires, separations, and earnings by worker group	`censusapi`	Yes
LEHD Origin-Destination (LODES)	Where people live and work, by census block	`lehdr`	No

Census Bureau: trade, indicators, and geography

Source	Economists use it for	R	Key
International Trade	Monthly imports and exports by product and partner	`censusapi`	Yes
Economic Indicators	Retail sales, housing starts, and manufacturing	`censusapi`	Yes
TIGER/Line boundaries	Tract, county, and ZIP code tabulation area maps	`tigris`	No
Census Geocoder	Assigning addresses to tracts and blocks	`tidygeocoder`	No

Geographic identifiers and boundaries link nearly every source in this catalog

Federal statistics: prices, income, and labor

Source	Economists use it for	R	Key
Bureau of Labor Statistics	CPI, PPI, employment, and wages (QCEW, LAUS, OEWS)	`blsR`	Yes
Bureau of Economic Analysis	GDP and personal income by state and county	`bea.R`	Yes
FRED and ALFRED	Macro and regional series, including past vintages	`fredr`	Yes
IPUMS	Harmonized Census, ACS, and CPS microdata; NHGIS tables	`ipumsr`	Yes
HUD User	USPS ZIP code crosswalks and Fair Market Rents	`httr2`	Yes

BLS | BEA | FRED | IPUMS

Federal statistics: energy, finance, and spending

Source	Economists use it for	R	Key
Energy Information Administration	Energy prices, production, and consumption by state	`eia`	Yes
CFPB HMDA	Mortgage applications and lending decisions	`httr2`	No
USAspending	Federal grants, contracts, and payments by place	`httr2`	No
Treasury Fiscal Data	Federal debt, receipts, and interest rates	`httr2`	No
FDIC BankFind	Bank branches and financial reports	`httr2`	No

EIA Open Data

USDA: production and markets

Source	Economists use it for	R	Key
NASS Quick Stats	Acreage, yields, prices, inventories, and the Census of Agriculture	`rnassqs`	Yes
NASS Cropland Data Layer	Annual 30-meter crop and land-cover map	`CropScapeR`	No
AMS Market News	Cash grain, livestock, dairy, and specialty crop prices	`httr2`	Yes
FAS Open Data	Global supply and use (PSD), trade (GATS), export sales	`httr2`	Varies
ERS ARMS	Farm finances, costs of production, and practices	`httr2`	api.data.gov

Quick Stats API | MyMarketNews | ERS data APIs

USDA: food, soils, snow, and forests

Source	Economists use it for	R	Key
FoodData Central	Nutrient composition of foods	`httr2`	api.data.gov
NRCS Soil Data Access	Soil properties and productivity (SSURGO)	`soilDB`	No
NRCS Air and Water Database	Snowpack and soil moisture (SNOTEL, SCAN)	`soilDB`	No
Forest Inventory and Analysis	Forest area, volume, growth, and removals	`rFIA`	No

Soil Data Access

Environment: water, drought, and hazards

Source	Economists use it for	R	Key
USGS Water Data	Streamflow and groundwater levels	`dataRetrieval`	No
Water Quality Portal	Samples from federal, state, and tribal monitoring	`dataRetrieval`	No
U.S. Drought Monitor	Weekly drought categories by county	`httr2`	No
OpenFEMA	Disaster declarations, flood insurance claims, and assistance	`httr2`	No

USGS Water Data APIs | Drought Monitor services | OpenFEMA

Environment: weather and climate

Source	Economists use it for	R	Key
NOAA NCEI	Daily station weather records (GHCN-Daily)	`httr2`	No
PRISM	Gridded U.S. temperature and precipitation	`prism`	No
gridMET and Daymet	Daily gridded weather for exposure measures	`climateR`, `daymetr`	No
NASA POWER	Global daily weather and solar radiation	`nasapower`	No

NCEI data service | nasapower

Environment: land, pollution, and energy

Source	Economists use it for	R	Key
NASA Earthdata	MODIS and Landsat imagery: vegetation, land cover, fire	`earthdatalogin`, `rstac`	Account
Google Earth Engine	Processing large satellite archives	`rgee`	Account
EPA Air Quality System	Monitor-level air pollution	`RAQSAPI`	Yes
EPA ECHO	Facility permits, inspections, and enforcement	`echor`	No
EPA Envirofacts	Toxic releases (TRI) and greenhouse gas reporting	`httr2`	No
NREL developer APIs	Solar resource, utility rates, fueling stations	`httr2`	api.data.gov

International

Source	Economists use it for	R	Key
World Bank Indicators	Development indicators by country and year	`WDI`	No
FAOSTAT	Agricultural production, trade, prices, and land use	`FAOSTAT`	Varies
UN Comtrade	Bilateral trade by product	`comtradr`	Yes
OECD Data Explorer	Country statistics, including agricultural support	`rsdmx`	No
IMF	Financial statistics and commodity prices	`imf.data`	No
Eurostat	EU regional and sectoral statistics	`eurostat`	No

Not every source has an API

Bulk download or license

- RMA Summary of Business: crop insurance by county and crop
- IRS Statistics of Income: income by ZIP code and county migration
- ERS data products such as the Food Environment Atlas
- State agencies: water rights, land records, and licensing
- Commercial data such as scanner or point-of-sale records, under a license

A download script records the URL, the file, and the download date

APIs change and disappear

- NOAA moved station data from the Climate Data Online API to a new data service, and the `rnoaa` package left CRAN in 2024
- EPA removed EJScreen and its map services in February 2025; reconstructions are now hosted outside EPA
- USGS plans to retire the services behind the `readNWIS` functions in 2027

Save the raw response and the retrieval date so the data can be reproduced after a service changes

Today’s four examples

Catalog group	Source	Returned to R
USDA	Cropland Data Layer	Raster map or summary table
Environment: water	USGS Water Data	Daily time series
Environment: weather	NASA POWER	Daily weather table
Census Bureau	American Community Survey	Estimates, margins of error, and map geometry

The four sources differ in access, unit of observation, and geography. Each example begins with a small request that can be inspected before the request is expanded

Run R in the HTML slides

- Wait for the R ready message after opening the HTML lecture
- Select **Run Code** to execute a live cell
- Change a named value such as `threshold` or `selected`
- Select **Start Over** to restore the original code

The API retrieval slides show the complete request code for RStudio. The live cells use compact classroom extracts shaped like the returned records, so every student can run them from the course site

The PDF contains the initial code for reference

Install the R packages

Run this installation once on your computer

```r
install.packages(c(
  "CropScapeR",
  "dataRetrieval",
  "nasapower",
  "tidycensus",
  "ggplot2",
  "dplyr"
))
```

Create a folder for files returned by the services

```r
dir.create("data/raw", recursive = TRUE, showWarnings = FALSE)
```

Follow-along repository

github.com/laurenchenarides/arec_570_demo

- Today's four requests are in `api_examples.R` at the `5.2-retrieve` tag
- The full API catalog is in `docs/api-catalog.md`
- Lectures 5.3 to 5.5 turn the same project into a reproducible, documented repository, one tagged stage per lecture

A retrieval workflow

1. Read the source documentation
2. Define the place, time period, variables, and unit of observation
3. Run a small request
4. Inspect column names, values, units, and missingness
5. Save the returned file or object
6. Record the request and retrieval date
7. Expand the request after the checks are complete

Example 1: land use

Question

Which crop and land-cover classes does the Cropland Data Layer assign in Larimer County in 2024?

Source

USDA NASS Cropland Data Layer through the CropScape service

Observation

Each raster cell has a modeled crop or land-cover class for a particular year

CropScape documentation | CropScapeR documentation

Request a Cropland Data Layer map

```
library(CropScapeR)

larimer_cdl <- GetCDLData(
  aoi = "08069",
  year = 2024,
  type = "f",
  save_path = "data/raw/cdl_larimer_2024.tif"
)

raster::plot(larimer_cdl)
```

`08069` is the five-digit Census code for Larimer County

The saved GeoTIFF contains a grid of land-cover codes and the information needed to locate those cells on a map

Use the land-use data

- Map crop locations
- Calculate acreage by crop class
- Compare annual crop patterns
- Define land-use exposure near water infrastructure or development

Try it: summarize land-cover records

Change `selected`, then run the cell. The classroom extract has the same class-and-acreage structure used in a CDL summary

R Code | Start Over | Run Code

```r
cdl <- data.frame(
  class = c("Corn", "Alfalfa", "Grassland", "Developed", "Other"),
  acres = c(14200, 8900, 180500, 72400, 12600)
)
selected <- c("Corn", "Alfalfa")
chosen <- subset(cdl, class %in% selected)
chosen$county_share <- round(chosen$acres / sum(cdl$acres), 3)
chosen
```

Assess the land-use data

- Crop-class definitions for each year
- Classification accuracy for the study region and crop
- Pixel resolution and coordinate system
- Treatment of double cropping and small fields
- Consistency of the study boundary across years

The CDL records modeled land-cover classes. Parcel ownership, zoning, and management practices require additional sources

Example 2: water

Question

What was the daily flow of the Cache la Poudre River during the 2024 growing season?

Source

USGS Water Data API

Observation

Each row reports a daily statistic for one parameter at one monitoring location

USGS monitoring location 06752260 | dataRetrieval tutorial

Request daily streamflow

```
1  library(dataRetrieval)
2
3  poudre_flow <- read_waterdata_daily(
4    monitoring_location_id = "USGS-06752260",
5    parameter_code = "00060",
6    time = c("2024-04-01", "2024-09-30")
7  )
8
9  names(poudre_flow)
10 head(poudre_flow)
```

`00060` is the USGS parameter code for discharge

The returned columns include date, value, unit, approval status, qualifiers, station geometry, and a time-series identifier

Try it: define a low-flow threshold

Change `threshold`, then run the cell. The compact series represents the date and value columns returned by the water request

R Code | Start Over | Run Code

```r
poudre_flow <- data.frame(
  date = as.Date("2024-07-01") + 0:13,
  flow = c(970, 910, 820, 760, 680, 610, 540, 490, 450, 420, 390, 360, 335, 310)
)
threshold <- 500
cat("Days below threshold:", sum(poudre_flow$flow < threshold))
plot(flow ~ date, poudre_flow, type = "l", xlab = "", ylab = "Daily discharge")
abline(h = threshold, col = "#C4513A", lty = 2)
```

Assess the streamflow series

- Confirm the monitoring location and parameter
- Read the unit attached to each series
- Examine provisional and approved status
- Identify missing dates and changes in the operating record
- Learn whether reservoirs, diversions, or return flows affect the gauge
- Define the geographic relationship between the gauge and the study area

The gauge measures conditions at one point on a river network. A study of irrigation access may also require diversion service areas, water rights, or delivery records

Example 3: weather

Question

What temperature and precipitation conditions occurred near Fort Collins during the 2024 growing season?

Source

NASA Prediction Of Worldwide Energy Resources or POWER

Observation

Each row reports modeled weather conditions for a grid cell and day

NASA POWER documentation | nasapower documentation

Request daily weather

```
1  library(nasapower)
2
3  fort_collins_weather <- get_power(
4    community = "AG",
5    lonlat = c(-105.08, 40.58),
6    pars = c("T2M_MAX", "T2M_MIN", "PRECTOTCORR"),
7    temporal_api = "daily",
8    dates = c("2024-04-01", "2024-09-30")
9  )
10
11 head(fort_collins_weather)
```

`AG` selects parameters prepared for agricultural applications

The three requested measures are maximum temperature, minimum temperature, and precipitation

Use the weather data

- Count high-temperature days
- Calculate growing degree days
- Sum growing-season precipitation
- Construct weather exposures for farms, counties, or markets

Try it: define weather exposures

Change `heat_threshold`, then run the cell. The extract contains daily maximum temperature and precipitation

R Code | Start Over | Run Code

```r
tmax <- c(29, 31, 33, 35, 36, 34, 30, 28, 32, 37, 38, 35, 31, 29)
precip <- c(0, 0, 1.2, 0, 0, 4.8, 0, 7.1, 0, 0, 0, 0.5, 2.4, 0)
heat_threshold <- 35
c(heat_days = sum(tmax >= heat_threshold),
  total_precip_mm = sum(precip))
plot(tmax, type = "b", xlab = "Day", ylab = "Maximum temperature (C)")
abline(h = heat_threshold, col = "#C4513A", lty = 2)
```

Assess the weather data

- Grid-cell resolution
- Coordinates assigned to each study unit
- Parameter definition and unit
- Time standard and daily aggregation
- Version and revision status

POWER provides grid-cell estimates. Conditions at fields within the same cell can vary

Example 4: Census ACS

Question

How does median household income vary across census tracts in Larimer County?

Source

American Community Survey five-year estimates

Observation

Each row reports an estimate and margin of error for one variable and geographic area

Census API guidance | tidycensus documentation

Get a Census API key

The Census Bureau requires a key for every request

1. Complete the form at api.census.gov/data/key_signup.html
2. Open the email from the Census Bureau and select the activation link
3. Keep the 40-character key from that email for the next step

Store the Census API key

Open the user-level environment file from R

```
install.packages("usethis")
usethis::edit_r_environ()
```

Add one line, save the file, and restart R (**Session > Restart R**)

```
CENSUS_API_KEY=paste_your_key_here
```

Check that R can read it: `nzchar(Sys.getenv("CENSUS_API_KEY"))` returns `TRUE`

The file stays in your home directory, outside every project. Scripts read the key with `Sys.getenv("CENSUS_API_KEY")`

Request tract estimates and geometry

```r
library(tidycensus)

larimer_income <- get_acs(
  geography = "tract",
  variables = c(median_income = "B19013_001"),
  state = "CO",
  county = "Larimer",
  year = 2024,
  survey = "acs5",
  geometry = TRUE,
  key = Sys.getenv("CENSUS_API_KEY")
)

head(larimer_income)
```

The result contains a geographic identifier, name, estimate, margin of error, and tract geometry

Map the ACS estimates

```
1  library(ggplot2)
2
3  ggplot(larimer_income) +
4    geom_sf(aes(fill = estimate), color = NA) +
5    labs(
6      fill = "Median income",
7      title = "Median household income by census tract"
8    )
```

The default margin of error describes a 90 percent confidence interval

$$\text{estimate} \pm \text{margin of error}$$

Try it: use the ACS margin of error

Change `relative_moe_limit`, then run the cell. The classroom extract contains one estimate and margin of error for each tract

R Code | Start Over | Run Code

```r
acs <- data.frame(
  tract = c("001", "002", "003", "004"),
  estimate = c(68400, 52100, 79300, 44700),
  moe = c(4100, 8700, 5200, 9600)
)
relative_moe_limit <- 0.15
acs$relative_moe <- round(acs$moe / acs$estimate, 2)
acs$retain <- acs$relative_moe <= relative_moe_limit
acs
```

Assess the ACS estimates

- Treat the five-year estimate as a pooled period ending in the named year
- Retain the margin of error
- Identify estimates with large uncertainty
- Use geographic identifiers as character values
- Record the tract boundary vintage
- Adjust monetary values when comparing periods

ACS data can describe the population and socioeconomic setting for a land or water study. Exposure measures and program participation may come from separate sources

The unit changes across sources

Source	Unit returned by the request	Time	Geography
Cropland Data Layer	Raster cell	Annual	Grid
USGS Water Data	Gauge and day	Daily	Monitoring point
NASA POWER	Grid cell and day	Daily	Weather grid
Census ACS	Estimate and area	Five-year period	Census tract

A combined dataset requires a written rule for assigning pixels, gauges, weather cells, and census areas to the study units

Combining the downloads

For every join record

- Identifier used by each source
- Geographic crosswalk or spatial rule
- Time aggregation rule
- Number of observations before and after the join
- Unmatched locations and periods
- Treatment of revised, suppressed, and missing values

The combined analytic sample includes the observations that satisfy all source coverage and matching rules

The reading: allocating the collection budget

Dominitz and Manski ask how to allocate a fixed budget across collection processes that differ in cost and information (Dominitz and Manski 2017)

API access reduces the time required to retrieve additional observations

Research time is also required to

- Verify definitions and units
- Examine coverage, missing values, flags, and uncertainty
- Select spatial and temporal aggregation rules
- Validate joins across sources
- Archive metadata and retrieval records

Question: How should a researcher allocate one additional week across these tasks?

Select the access method

Project need	Useful access method
A defined subset repeated across places or periods	API request
Most records in a fixed release	Bulk download
A few values needed once	Web table or manual download
Repeated data cleaning and analysis	Scripted retrieval with archived raw files

Choose the method after reviewing the source documentation update schedule and request limits

Document the retrieval

Keep these items with the project

- Source and dataset name
- Data documentation
- Retrieval script
- Parameters submitted through the script
- Retrieval date
- Raw files returned by the service
- Processing script and data dictionary
- Package and R versions
- Validation results

Store API keys in the user environment file and exclude that file from version control

Class share: where are you in the process?

We will go around the room. For your project describe

1. The outcome, exposure, or behavior you need to observe
2. The primary, secondary, or combined collection approach you expect to use
3. The two catalog sources you marked, or another source or tool you have found helpful
4. The part of the collection process you are still working through
5. The next concrete step in your data search or collection

Fork or template?

Two ways to copy a GitHub repository into your account

	Fork	**Template**
History	Keeps every commit (and tags, if you uncheck *Copy the main branch only*)	Starts a new history
Link to the original	Pull later updates; propose changes back	None
Visibility	Public when the original is public	Public or private
Use it for	Following a project that keeps changing	Starting your own project from a standard layout

For this course, **fork** the demo repository: the fork keeps the tag for each lecture, and **Sync fork** brings in updates. Keep your own research in a separate private repository

Before Lecture 5.3

Complete Parts One through Three of docs/setup-rstudio.md

1. Install Git and restart RStudio
2. Create a GitHub account and connect RStudio with `usethis` and a personal access token
3. Request, activate, and store a Census API key
4. Fork the demo repository (uncheck *Copy the main branch only*) and open it in RStudio as a new project

Bring a laptop where `usethis::git_sitrep()` runs without errors and `nzchar(Sys.getenv("CENSUS_API_KEY"))` returns `TRUE`

Summary

APIs provide scripted access to secondary data

R packages translate requests and responses into familiar R objects

Source documentation defines the population, measures, timing, geography, and uncertainty

Small requests make the returned structure easier to inspect

Retrieval records support reproducibility

Data joins help define the final analytic sample

References

Dominitz, Jeff, and Charles F. Manski. 2017. “More Data or Better Data? A Statistical Decision Problem.” *The Review of Economic Studies* 84 (4): 1583–605. https://doi.org/10.1093/restud/rdx005.

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